

# Week 3: Cloud Computing Models and Services

Cloud Computing Models and Services

# Lecture Outline

- Introduction
- Characteristics of Cloud Computing
- Cloud Deployment Models
  - Public Cloud
  - Private Cloud
  - Community Cloud
  - Hybrid Cloud

# Introduction

- This lecture will help to understand the architectural model of cloud computing.
- The standard definitions and globally accepted NIST (National Institute of Standards and Technology) was used to establish a baseline.
- NIST identified five characteristics.
- Understanding the cloud deployment models is important to make informed decisions.
- This will help to pick appropriate cloud solution to satisfy personal or business needs.

# Introduction (Contd.)

- The NIST model of cloud computing separates cloud computing into two categories.
  - Based on the operational or deployment pattern of the cloud.
  - Based on the nature of service that the cloud provides.
- Cloud Modelling based on Deployment:
  - Focus on access boundary and location of the cloud establishment
- Cloud Modelling based on service delivery:
  - Describes the type of computing service offered to the users by the service provider

# Standard Cloud Model

- Computing technology has relied on few institutions and organizations for standardization and guidelines since its inception.
- Those institutions included:
  - The NIST of United States.
  - The International Organization for Standardization (ISO).
  - The European Telecommunications Standards Institute (ETSI).
- Others are Cloud Security Alliance (CSA).
  - To publish guidelines focusing to set up some standard model for cloud computing.
- The most acknowledged is the NIST Model.

# NIST Model

- NIST model is comprised of:
  - Five essential characteristics
  - Three service models, and
  - Four deployment models.
- Points to note from the definition:
  - Cloud computing is a model and not a technology.
  - Cloud computing enables the users' access pools of computing resources via network.
  - The resources are shared among users and made available on-demand.
  - The prime benefit is the ease of use with very little management tensions for the users.

# Characteristics of Cloud Computing

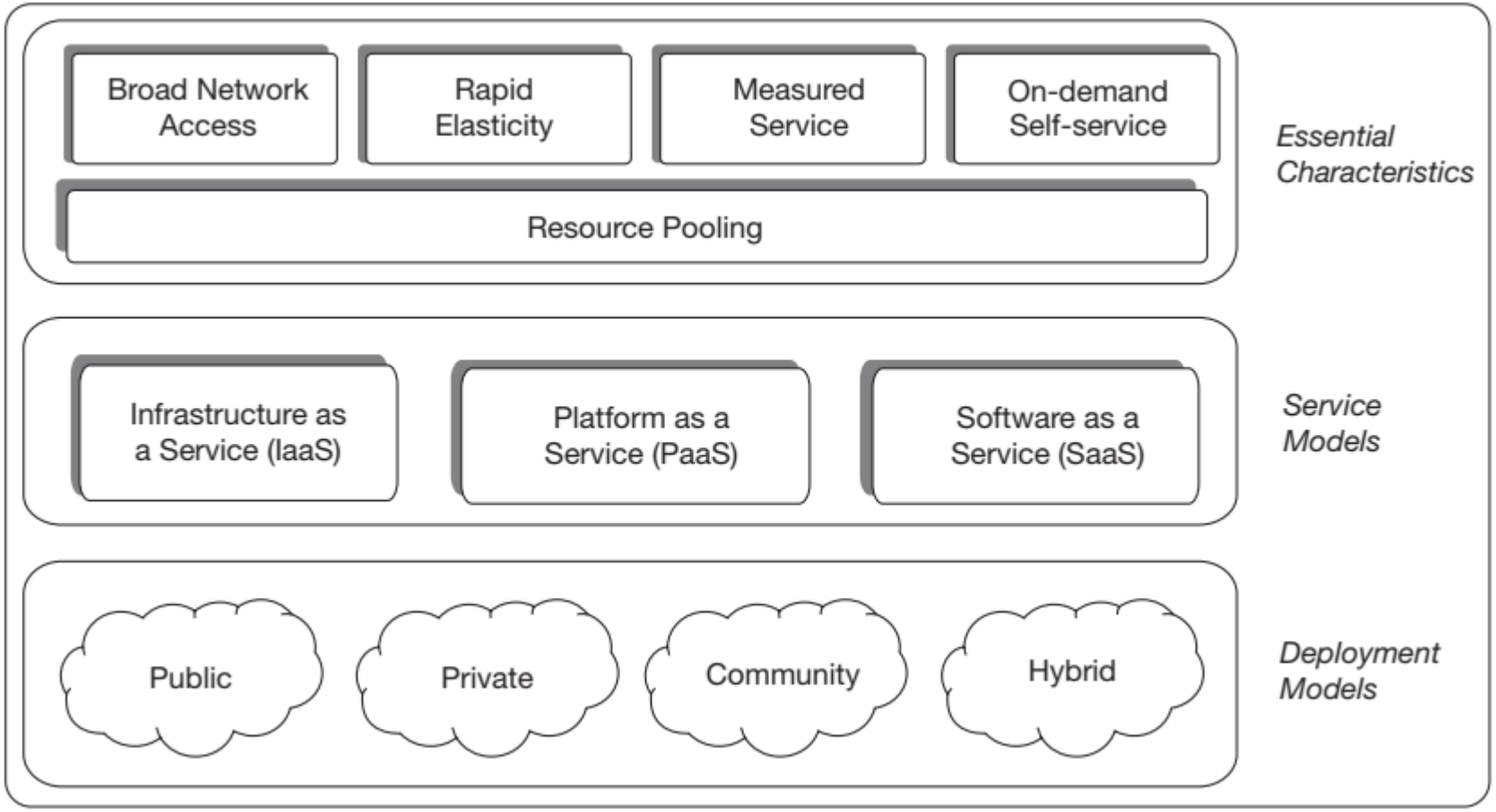



**Figure 3:** NIST Cloud Computing Model

# Cloud Deployment Models

- Cloud services can be arranged or deployed in several ways.
- The deployment choice depends on the requirements of the consumer organization.
- The deployment model:
  - describes the utility of a cloud
  - specifies its access boundary
  - indicates the relative location of the cloud concerning the location of a consumer

# Cloud Deployment Models (Contd.)

- There are four categories of cloud deployment:
  - Public Cloud
  - Private Cloud
  - Community Cloud
  - Hybrid Cloud

# Cloud Deployment Models (Contd.)

- **Public Cloud**
  - Public cloud is a computing service offered by external providers through the internet as it is enabled for everyone who wants to buy or use it.
  - Physical location wise this cloud remains external or off-premises.
  - Consumers remotely access the service.
  - Public cloud is also referred as external cloud.
  - Public cloud shares one-to-many relationship with consumers.

# Cloud Deployment Models (Contd.)

- **Public Cloud**
  - It is hosted and managed by some computing vendor, who establishes datacenters to provide service.
  - Consumers under this cloud deployment model are entirely free from any headache of infrastructure administration and system management.
  - Consumers have low degree of control over the cloud.
  - Cloud service provider manages the service whereas the organization uses them.

# Cloud Deployment Models (Contd.)

- **Public Cloud**
  - Public cloud deployment promotes multi-tenancy at its highest degree.
  - This increases resource utilization rates and decreases vendor's cost of service delivery.
  - Through this model consumers can access potentially superior service at a lower cost.
  - Amazon Web Services, Google Cloud, Microsoft Azure and Salesforce.com are popular public clouds.

# Cloud Deployment Models (Contd.)

- **Examples of Public Cloud Providers**
  - Amazon Web Services
    - a secure cloud platform, offering database storage services, delivery of content, and other functionalities so that companies can grow.
  - Google Cloud
    - With the Google Cloud, You can modernize workloads with one of the best infrastructures in the world and streamline migrations with pre-packaged cloud infrastructure solutions for SAP, VMware, Windows, Oracle, data center migrations and other business workloads

# Cloud Deployment Models (Contd.)

- **Examples of Public Cloud Providers**
  - Microsoft Azure and Salesforce.com are popular public clouds.
    - Azure, Microsoft's cloud platform, provides a broad set of IaaS and PaaS services that enable companies to move their business applications to the cloud or develop new applications natively with the highest levels of security and flexibility.

# Cloud Deployment Models (Contd.)

- **Private Cloud**
  - A private cloud is a computing model that offers a secure and dedicated environment for a single business.
  - Private clouds can be hosted within an organization's data center, at a third-party co-location facility, or by a private cloud provider.
  - This cloud computing model is used exclusively by a particular organization and never shared with other organizations.
  - This deployment does not provide open access to all.
  - Private cloud is also referred to as internal cloud.

# Cloud Deployment Models (Contd.)

- **Private Cloud**
  - Private cloud is user for high security and critical systems.
  - Private cloud may reside either inside consumer organization's premises (on-premises) or outside (off-premises) at any neutral location.
  - Private cloud shares one-to-one relationship with consumer.




**Figure 5:** On-premises Private Cloud

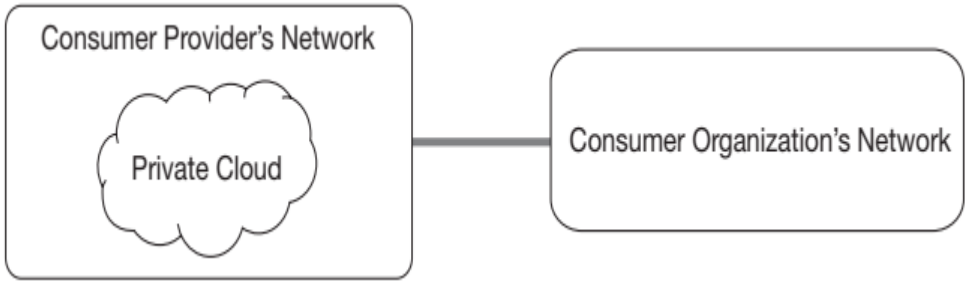



**Figure 6:** Off-premises Private Cloud

# Cloud Deployment Models (Contd.)

- **Private Cloud**
  - The feature of multi-tenancy does not apply in private cloud.
  - Consumers have more control over the cloud.
  - The cost of availing private cloud is higher than public cloud.
  - The organization have their administrators to manage private cloud services.
  - Private Cloud providers: VMWare, Dell, Oracle, CISCO

# Cloud Deployment Models (Contd.)

- **Examples of Private Cloud Service Providers**
  - **AWS:**
    - AWS Virtual Private Cloud services allow you to manage your virtual networking environment, including resource placement, connectivity, and security.
  - **Google Private Cloud:**
    - Google Private Cloud can auto-configure your virtual topology, prefix ranges for your subnets, and network policies, or you may configure them yourself.
    - Communication between organizations can be configured privately without bandwidth bottlenecks or single points of failure.
  - **Dell:**
    - Dell's Microsoft Hyper-VCloud Fast Track program is a comprehensive approach to cloud computing.

# Cloud Deployment Models (Contd.)

## Private Cloud Vs Public Cloud

| Private Cloud | Public Cloud |
|---|---|
| Can be both of type on-premises or off-premises. | There can't be any on-premises public cloud deployment. |
| On-premises private cloud can be delivered over private network. | Can only be delivered over public network. |
| Does not support multi-tenancy for unrelated and external tenants. | Demonstrates multi-tenancy capability with its full ability. |
| Resources are for exclusive use of one consumer (generally an organization). | Resources are shared among multiple consumers. |
| A private cloud is accessible to a restricted number of people. | This facility is accessible to anyone. |

# Cloud Deployment Models (Contd.)

## Private Cloud Vs Public Cloud

| Private Cloud | Public Cloud |
|---|---|
| This is for organizational use. | Can be used both by organization and user. |
| Consumers have important roles to play in management of the cloud. | Management is entirely provider's responsibility. |
| Consumers have more control over the environment. | Consumers have very less control or no control. |
| Provides more confidence regarding data security as remains under the control of consumer organization's security boundary. | Public cloud deployment often creates concerns regarding security and privacy of data. |
| Cost of computing is more in comparison to public cloud. | This is more economical, as multiple unrelated consumers (tenants) share same infrastructure. |

# Cloud Deployment Models (Contd.)

## Private Cloud Vs Public Cloud

| Private Cloud | Public Cloud |
|---|---|
| Not ideal for promoting pay-as-you-use philosophy. | Ideal model for practicing pay-as-you-use philosophy. |
| Resource may often remain idle, resulting resource wastage. | Resource utilization is optimum, due to presence of larger number of consumers. |
| Not so environment friendly like public cloud. | Promotes green computing at its best. |

# Cloud Deployment Models (Contd.)

- **Community Cloud**
  - A cloud-based infrastructure that enables multiple organizations to share services and resources derived from common regulatory and operational requirements.
  - It offers a shared platform and resources for different organizations to work on their business requirements.
  - It is operated and managed by community members, third-party vendors, or even both.

# Cloud Deployment Models (Contd.)

- **Community Cloud**
  - This deployment model allows access to several organizations or consumers – belonging to a community.
  - The model is built to serve some common and specific purpose.
  - This model allows the sharing of infrastructure and resources among multiple consumers belonging to a single community.

# Cloud Deployment Models (Contd.)

- **Community Cloud**
  - Community cloud deployment can be on-premises or off-premises.
  - It may be identified as a generalized form of private cloud.
  - It is cheaper compared to private cloud.
  - Community cloud provides advantages of public cloud deployment, combined with the promise of private cloud like security and privacy of data.

# Cloud Deployment Models (Contd.)

- **Community Cloud**
  - Cloud providers have developed Community Cloud offerings, and some organizations are already seeing the benefits.
  - Scenarios of the Community Cloud model
    - Educational institutions:
      - The community cloud model is the most suited model for educational institutions as it allows them to share information, research material, and educational content on the cloud.
      - In group projects, the model can be used to facilitate question-and-answer sessions that help foster collaboration.
    - Health care industry:
      - The community cloud model has use cases for the healthcare sector. This sector deals with highly sensitive information when collaborating with several pharmaceutical companies.

# Cloud Deployment Models (Contd.)

**Private Cloud Vs Community Cloud**

| Private Cloud | Community Cloud |
|---|---|
| One private cloud can be used by one consumer (may be an organization). | One community cloud can be used by one community. |
| Access is restricted among members of a single consumer. | Access is restricted among members of a single community. |
| Resources of a private cloud are for exclusive use of one consumer. | Resources are shared among multiple consumers of a single community. |
| May reside at consumer's premises. | May reside at some consumer's premises within the community. |

# Cloud Deployment Models (Contd.)

## Private Cloud Vs Community Cloud

| Private Cloud | Community Cloud |
| --- | --- |
| Private cloud (on-premises) can be delivered over private network. | To provide access to multiple consumers (of a community), delivered over public network. |
| Does not support multi-tenancy. | Multi-tenancy is supported. |
| This is for organizational use. | This one is for use of communities. |

# Cloud Deployment Models (Contd.)

- **Hybrid Cloud**
  - It is created by combining private or community deployment with public cloud deployment together.
  - The organization operates the private cloud whereas cloud service providers manage the public cloud.
  - This model helps businesses to take advantage of private or community cloud by storing critical applications and data there.
  - It also provides cost benefit by keeping shared data and applications on the public cloud.

# Cloud Deployment Models (Contd.)

- **Hybrid Cloud**
  - Example:
  - For a project, NASA is using the Hybrid cloud computing deployment
  - Its open-source cloud-registered project NEBULA uses a private cloud for research and development and in addition a public cloud to share datasets with external partners and the public.

# Cloud Deployment Models (Contd.)

- **Hybrid Cloud**

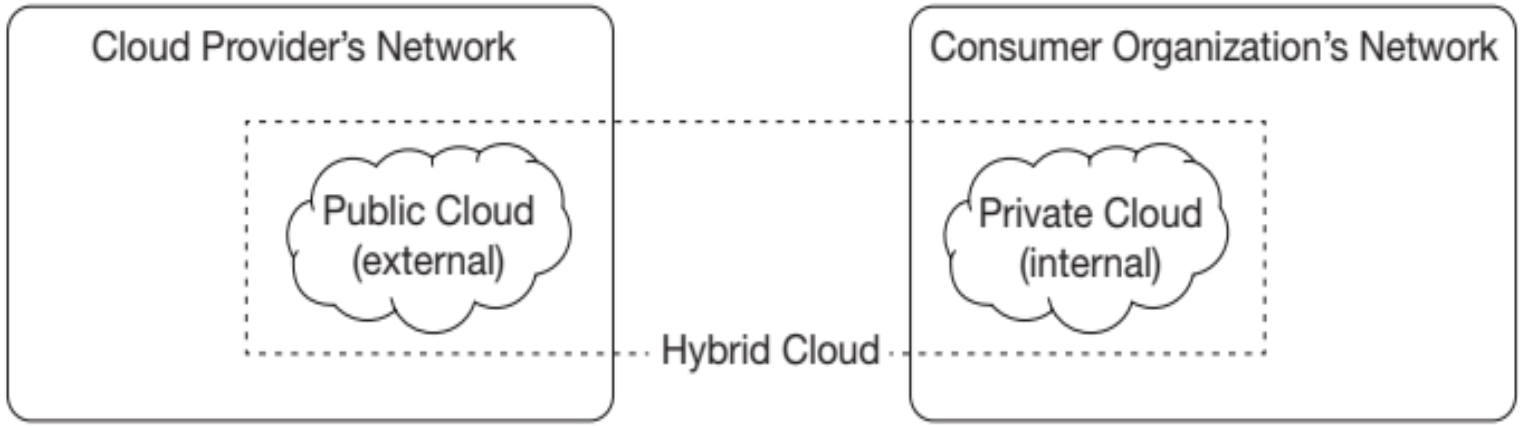



**Figure 7:** Hybrid Cloud Model

# Cloud Deployment Models (Contd.)

- Cloud deployment states how exclusively a cloud consumer can use computing resources.
- To select the appropriate cloud deployment model, one will have to assess:
  - the business needs
  - strengths and weaknesses of a model,
  - privacy/security requirements, and
  - goals of the organization.
- Hybrid Cloud service providers include Amazon, Hp, Google, Microsoft.

# Cloud Deployment Models (Contd.)

- **Economies of Scale**

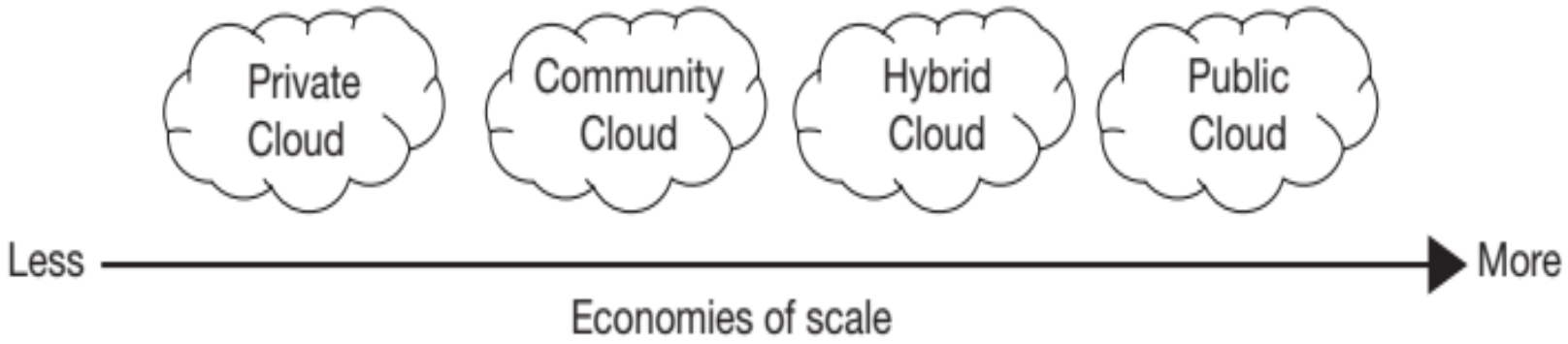



**Figure 8:** Variations of cost-effectiveness with different cloud deployments

# Cloud Deployment Models (Contd.)

- **Consumer's Authority**

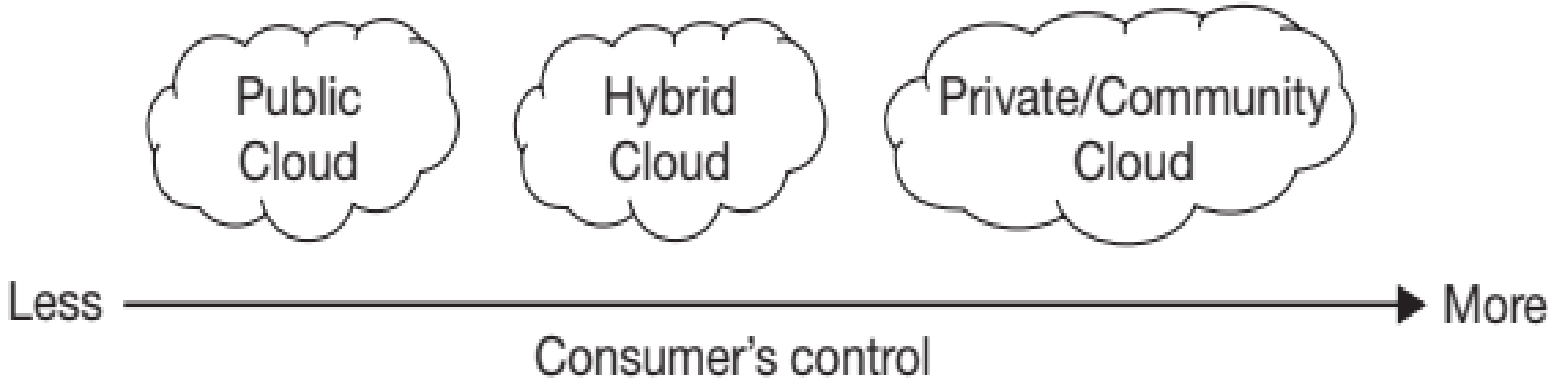



**Figure 9:** Variation in user's control over different cloud deployments

# Thank You